Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
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Project Title:	Gravitational Impacts on the Cell Cycle		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	32901	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
Task Description:	A new cell culture system has been developed specifically for the evaluation of environmental impacts on diverse aspects of the cell cycle of lympho-hematopoietic cells. In this culture system, large quantities of synchronous, newborn cells are generated automatically from cultures of immobilized cells undergoing long-term steady-state growth. The newborn cells are shed continuously from the culture vessels, which can be held stationary or rotated about the horizontal or vertical axes, and used for the cell cycle analyses. The first aim of the study is to assay the effects of gravity averaging on cell cycle phase durations, segregation of components between daughter cells, gene expression, and division plane orientations of mouse and human cell lines cultured in the vessels. The second aim will address the impacts of altered gravity on cell cycle aspects of the differentiation of the human myelomonocytic cell line U937. The third aim will employ the culture system to determine the basic properties of the cell cycle of activated normal human lymphocytes, and changes in the cycle during senescence and during growth in altered gravity. The key to all the studies is that the culture system permits cell cycle analyses on cells undergoing minimally disturbed steady-state growth so any		

	alterations detected in cell cycle properties will be due to altered gravity and not the procedures used to produce the synchronous cells or perform the cell cycle analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The culture technology to be employed in this project is designed to specifically address objectives of the space biology program, and to use NASA-developed technologies and research to expand understanding of fundamental cellular biological processes, as well as the mechanisms by which these processes sense and respond to the space environment. The project will address several of the emphasis areas for fiscal 2001 in cell and molecular biology, and developmental biology, including: the impact of gravity on cell cycle aspects of growth control, gene expression, mitosis, normal differentiation, and cellular senescence (life span). Furthermore, the work will be performed on hematopoietic cells, which have been the subject of considerable analysis of gravitational impacts, and which may be especially sensitive to the gravity vector. The goals will be accomplished using a technology for mammalian cell cycle research designed to model microgravity by gravity averaging. The very special property of this cell culture technique for cell cycle analyses is that the cells being examined are in minimally disturbed steady-state growth so that the fundamental properties of the cells growing in 1g, and the measurements of gravitational impacts, are not distorted or masked by uncontrolled stresses on the cells. To enhance the information content in cell cycle studies, other external influences on the cycle need to be minimized so that potential gravity effects can be maximized. No other procedure for cell cycle analyses can yield the breadth of information available with this technique. Furthermore, this project will introduce a new methodology to the field, and verification of its utility for cell cycle research and as a model for microgravity should stimulate widespread use of the technology and the cell cycle products produced, yielding a potential economic benefit from its development.
Task Progress:	The new rotating bioreactor for use in basic cell cycle research and for analyzing effects of gravity averaging on the cycle during steady-state growth has been significantly improved during the fiscal year. This culture system produces large quantities of synchronous, newborn cells automatically, from cultures of immobilized cells, for days. The newborn cells are shed continuously from the culture vessel while rotating about the horizontal axis and can be used for diverse cell cycle analyses. The culture system in the rotating bioreactor consisted of approximately 50 million cells attached to an adhesive-coated 100-sq.cm. nitrocellulose membrane filter held vertically in the vessel. The major new improvement related to the manner in which the cells were adhered to the surface. Previously, the cells were attached to membranes coated with either concanavalin A or poly-D-lysine outside the vessel and then the membrane was subsequently clamped into the bioreactor. In the improved system, the bioreactor was fully assembled with the coated membrane in place prior to attachment of the cells. The cells were then introduced by syringe through the exit port of the bioreactor in approximately 5 minutes. This approach was made possible by the placement of a flexible support screen behind the coated membrane to enable flushing the cells onto the membrane surface without distorting the flat membrane or introducing bubbles that could not be removed. This technique enabled rapid, aseptic initiation of each bioreactor experiment. The direction of culture medium flow was then reversed so that the medium flowed past the attached cells and out the exit port such that, at cell division, one new daughter cell was shed from the surface and exited the vessel while the other daughter remained attached to the surface. As a result of this growth process, the rotating vessel functioned as a continuous culture device, with no changes in the environment of the cells except for the rotation. Experiments in the bioreactor have been successfully performed with human MOLT-4 and mouse L1210 lymphocytic leukemia cells for up to nine generations, and essentially pure newborn cells, at concentration of approximately 10exp8 cells/day, were released throughout. The data from several experiments demonstrated that clinorotation had no effect on fundamental cell cycle properties, such as lengths of cell cycles, timing of initiation of S phase, lengths of G1 and G2, interdivision times, and cell volume distributions. To determine if these conclusions were applicable to other cell types, considerable effort was put into investigating the effects of gravity averaging on the CHO cell cycle with this technology. Since these fibroblast-like cells normally spread on surfaces, and as a consequence newborn cells are not normally shed from the surface during division, new procedures had to be developed in an attempt to prevent spreading. In collaboration with JRH Biosciences, several serum-free media formulations were tested for this purpose. It was found that spreading could be avoided with the various media, but in spite of this, both newborn cells consistently remained attached to the surface after division. It appeared that the division planes of the CHO cells were repeatedly perpendicular to the attachment surface and as a result both daughters remained in contact with the surface. This differed from the division process of lymphocytes in which division planes were rarely perpendicular to the attachment surface. Additional studies employing alternative cell lines in the bioreactor are now in progress.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Articles in Peer-reviewed Journals	Grover NB, Eward KL, Helmstetter CE. "Synchronous cultures from the baby machine: A model for animal cells." Mathematical Biosciences. 2004 Jul;190(1):87-96. PubMed PMID: 15172804 , Jul-2004
Patents	US Patent Application Serial No. 10/373,130. Pending, February 2003. Feb-2003 Helmstetter CE, Thornton M, Gonda S. "Bioreactor and methods for producing synchronous cells."
Presentation	Romero, A.L. "Cell cycle regulation in mouse leukemia cells." National Institutes of Health. Dec-2003
Presentation	Helmstetter, C.E. "New technology for cell cycle research." JRH Biosciences, Lexana, KS. Nov-2003
Presentation	Helmstetter, C.E. "Cell cycle studies with synchronous cultures." Midwest Research Institute, Palm Bay, FL. May-2004